RI048

Multidimensional Adaptive Probing of Agent Capability Boundaries

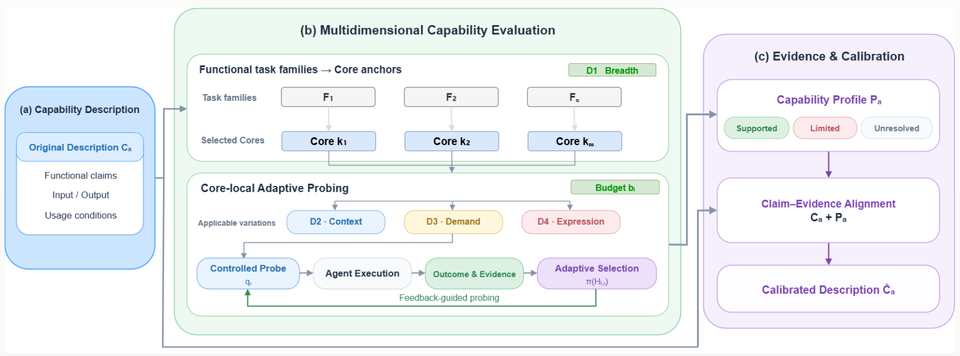
Quanda Wu[1*] · Yi Lin[2*] · Ke Li[3] · Qi Wang[4†]

[1] Beijing University of Posts and Telecommunications · [2] Institute of Scientific and Technical Information of China
[3] Kashgar Prefecture Electronic Information Industry Technology Research Institute · [4] National Institute of Metrology, China

Introduction

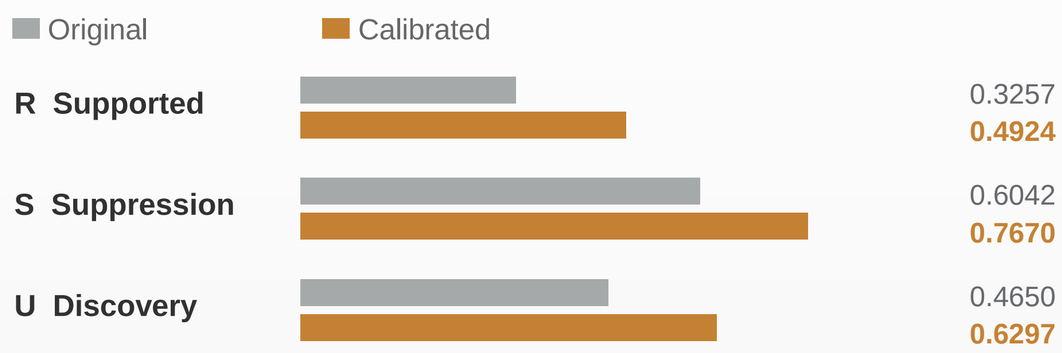
Capability descriptions guide agent discovery, yet may overstate supported functions or omit applicability conditions.

We use execution-grounded, adaptive probing to identify local capability boundaries and calibrate agent descriptions under a limited budget.



Results

Across 90 held-out queries (57 positive; 33 negative), calibrated descriptions improved discovery utility by 35.4% relative.

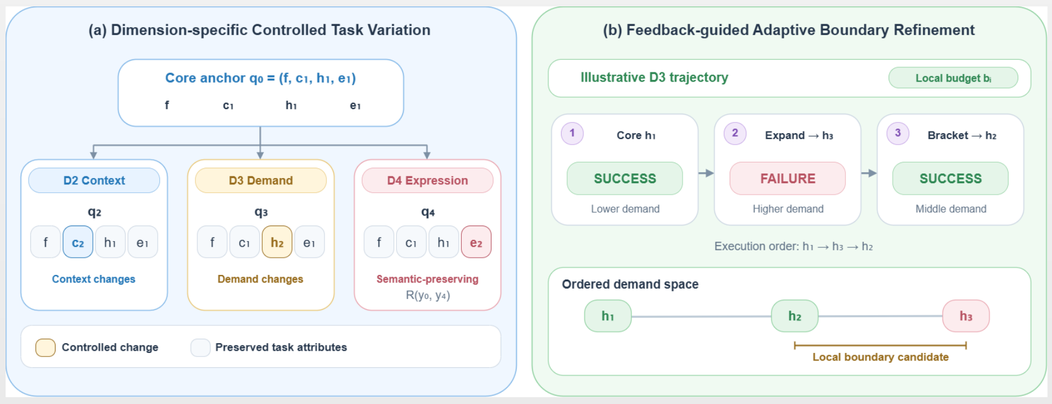


U: 0.4650 → 0.6297 | +0.1647 absolute

Objectives

Research goal Identify supported conditions, local limitations and unresolved capability regions with minimal execution cost.

Four perspectives Functional Breadth (D1) · Contextual Generality (D2) · Demand Depth (D3) · Semantic Robustness (D4).

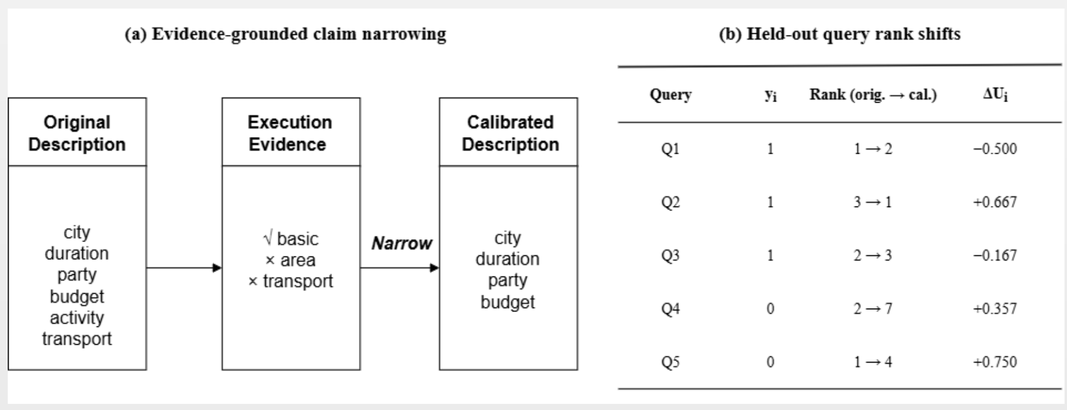


Feedback selects the next controlled probe around each core anchor.

Conclusion

Local execution evidence supports evidence-constrained expansion, narrowing and qualification of capability claims.

The calibrated descriptions improve discovery of supported targets and reduce exposure of unsupported targets.



Hotel recommendation case: ΔR = 0, ΔS = +0.5536.

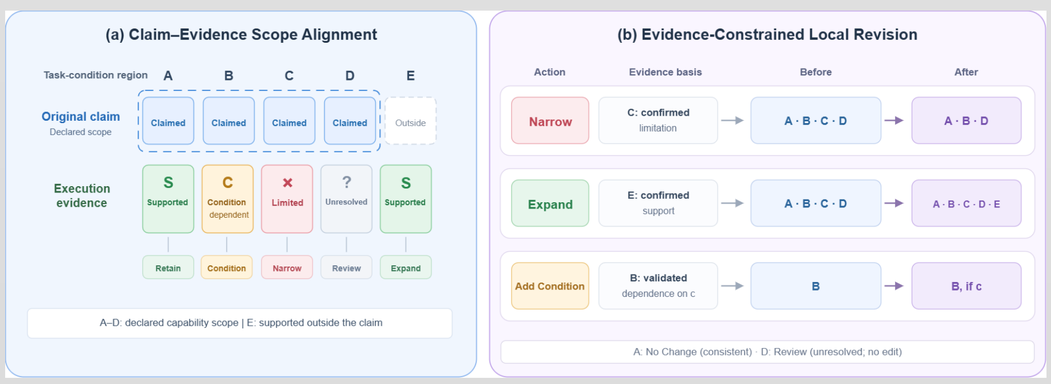
Materials & Methods

1 Construct representative core anchors from declared task families.

2 Expand, Contract, Contrast, Bracket or Resolve using execution feedback.

3 Verify task objectives, synthesize evidence and locally calibrate claims.

4 Compare original vs. calibrated descriptions with the same ACPs Discovery settings.



Experimental setting 15 agents · 18 Skills · 23 core anchors · 5 executions/anchor = 115 probing executions.

DeepSeek V4 Pro supports probe generation and description calibration. Held-out execution outcomes are manually reviewed against task objectives.

References

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Contact

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* Quanda Wu and Yi Lin contributed equally. † Qi Wang is the corresponding author.

Contact: wuquanda@bupt.edu.cn | wangqi@nim.ac.cn

Research setting: Wutong agent platform; ACPs Discovery; execution-grounded capability-description calibration.